Mapping Historical Mining Shafts Under Geometric Degeneracy: A Comparison of LiDAR-Only, LiDAR–Inertial, and Cable-Assisted Methods

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***Abstract*—Historical mining shafts are narrow, near-cylindrical excavations that are unsafe to inspect directly and prone to geometric degeneracy in LiDAR-based odometry. This paper compares five mapping strategies for cable-lowered inspection: FAST-LIO2, attitude- and cable-referenced scan concatenation, LiDAR-only KISS-ICP, cable-anchored KISS-ICP, and BIEVR-LIO. The anchored formulation combines an encoder-based prediction override, a soft axial constraint, and tilt regularization motivated by the suspended deployment geometry. Two field sequences are evaluated using manual depth measurements, cloud-to-cloud agreement, and registration repeatability. FAST-LIO2 failed to preserve the approximately axial shaft topology, while LiDAR-only KISS-ICP showed the largest axial contraction among the remaining methods. Across the two shafts, cable anchoring reduced KISS-ICP's mean absolute extent error from 0.473 to 0.153 m, a 68% reduction. Registration was least repeatable in axial translation and rotation about the shaft axis, consistent with the dominant near-cylindrical geometry. These results indicate that auxiliary motion cues can mitigate geometric degeneracy, although robustness depends strongly on how they are incorporated into the estimator.**

***Index Terms*—confined-environment mapping, LiDAR odometry, sensor fusion, geometric degeneracy, cable-lowered sensing**

I. Introduction

Historical mining shafts are hand-excavated vertical structures originally used to access gold-bearing veins in colonial mining regions. In the study region, these narrow, irregular excavations may reach depths of approximately 40 m, with diameters ranging from a few tens of centimeters to about 2 m. They are generally accessible only through a small surface collar and may connect to lateral galleries. Bare rock walls, absence of natural light, and uncertain structural stability make direct inspection unsafe. Nevertheless, three-dimensional documentation is relevant for heritage preservation and preliminary geotechnical assessment because abandoned shafts may remain hidden beneath vegetation, streets, or buildings [1], [2].

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Fig. 1: Historical mining shaft and field deployment setup. (left) Shaft collar partially hidden by vegetation, showing the narrow vertical access and the proposed device. (right) Example point-cloud of the shaft mapping mission.

Recent large-scale studies of subterranean robotics have established LiDAR-centric SLAM as a key technology for mapping and localization in GPS-denied underground environments [3]. Robotic systems have been deployed in hazardous confined environments such as planetary skylights, ventilation shafts, and underground mines [4]–[8]. These systems, however, generally rely on traversable surfaces, sufficient flight volume, or controlled interaction with the surrounding structure, conditions that may not be available in the narrow shafts considered here. Figure 1 shows the adopted alternative: a sensing payload is lowered from the surface by cable while operators remain outside the shaft.

This deployment creates a particularly challenging state-estimation problem. Because the payload operates within a few tens of centimeters of near-cylindrical rock walls, scans acquired at different depths can exhibit similar geometric associations. Translation along the shaft axis and rotation about it are therefore weakly constrained, whereas lateral motion and changes in shaft orientation are typically better constrained by the surrounding walls [9]. This anisotropic observability can lead to axial drift, map contraction, and unstable rotation about the shaft axis.

Auxiliary motion information can mitigate these weak geometric constraints. In our proposed system, the reel encoder provides an axial displacement cue, while the Mid-360 IMU supplies inertial data for the LiDAR–inertial and attitude-referenced configurations. Depending on the method, these cues are used for scan concatenation or registration, or omitted in the LiDAR-only formulation.

This work investigates how such auxiliary information affects mapping under shaft-axis degeneracy by comparing FAST-LIO2 [10], attitude- and cable-referenced scan concatenation, LiDAR-only KISS-ICP [11], cable-anchored KISS-ICP, and BIEVR-LIO [12] on identical recordings from two historical shafts. The main contributions are: 1) a comparison of LiDAR-only, LiDAR–inertial, and cable-assisted mapping methods under identical field conditions; 2) an anchored KISS-ICP formulation that combines encoder-based axial anchoring with tilt regularization motivated by the suspended deployment geometry; and 3) an evaluation of reconstructed axial extent, local geometric agreement, and registration sensitivity along the geometrically weak directions.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the payload, the instrumented cable, and the field workflow. Section IV develops the cable-encoder–anchored KISS-ICP formulation. Section V details the experimental setup, Section VI presents results and discussion, and Section VII concludes the paper.

II. Odometry in Geometrically Degenerate Environments

LiDAR odometry relies on the geometric constraints available for scan registration [9], [13]. KISS-ICP [11] provides a simple and robust LiDAR-only formulation based on point-to-point registration, whereas LiDAR–inertial methods introduce inertial measurements to complement geometric information. FAST-LIO2 [10] tightly couples both modalities through an iterated Kalman filter, while BIEVR-LIO [12] uses bump-image-enhanced voxel maps to improve robustness in geometrically challenging environments.

A fundamental limitation of geometry-based estimation arises when the observed structure does not sufficiently constrain all motion directions. Zhang *et al.* [9] characterize such degeneracy through the eigenstructure of optimization-based state estimation. Other approaches explicitly incorporate localizability into the estimation process. X-ICP [13], for example, identifies weakly constrained directions during LiDAR registration and regularizes them accordingly, while degeneracy-aware factors have also been introduced into graph-based SLAM formulations [14].

Weak observability can also be mitigated through auxiliary sensing: UWB ranging has been used in tunnel-like environments [15], while EKF-LOAM [16] combines LiDAR, wheel odometry, and inertial data in confined spaces. Cable-lowered inspection provides a deployment-specific alternative because tether displacement supplies an independent cue approximately aligned with the weak axial direction.

III. System Overview

The 4.24-kg sensing payload measures $(15 \times 27 \times 23)$ cm (width x length x height), reaching 31 cm in overall height when the Livox Mid-360 and its protective frame are included. It integrates a Livox Mid-360 LiDAR, an Intel RealSense D456 RGB-D camera, an Intel NUC 13 Pro Slim with an Intel Core i5 processor, LED illumination, and control electronics. Figure 2 shows the main hardware components.

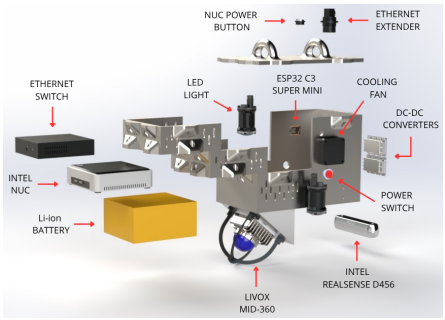


Fig. 2: Annotated internal layout of the sensing payload and its main hardware components.

The main payload components have an estimated cost of approximately US$6.65k, based on Brazilian retail prices collected in August 2026 and converted at BRL 5.18/USD. This estimate includes the Mid-360, D456, NUC, and approximately 765.5 g of PLA used for the 3D-printed structure. Auxiliary electronics, batteries, cabling, and the external deployment mechanism are not included.

The Mid-360 provides a $360^\circ \times 59^\circ$ field of view, a 0.1-m blind zone, and detection ranges of 40 m at 10% reflectivity and 70 m at 80% reflectivity [17]. Its built-in ICM40609 IMU supplies the inertial data used in the evaluated configurations. As shown in Fig. 3, the LiDAR is mounted beneath the payload at 90° to the deployment axis for lateral wall coverage. The D456 adds an $87^\circ \times 58^\circ$ depth field of view and RGB-D sensing under LED illumination [18].

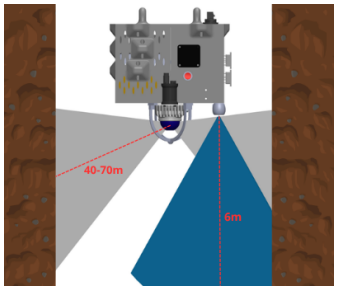


Fig. 3: Sensing geometry of the payload. The Mid-360 is mounted 90° to the deployment axis for lateral wall coverage, while the D456 provides shorter-range illuminated sensing.

The payload is lowered through a bearing-guided tripod using a Blue Robotics Fathom Slim ROV Tether. Reel displacement is measured by an E38S6G5-600BG24N optical encoder with 600 pulses per revolution and quadrature output.

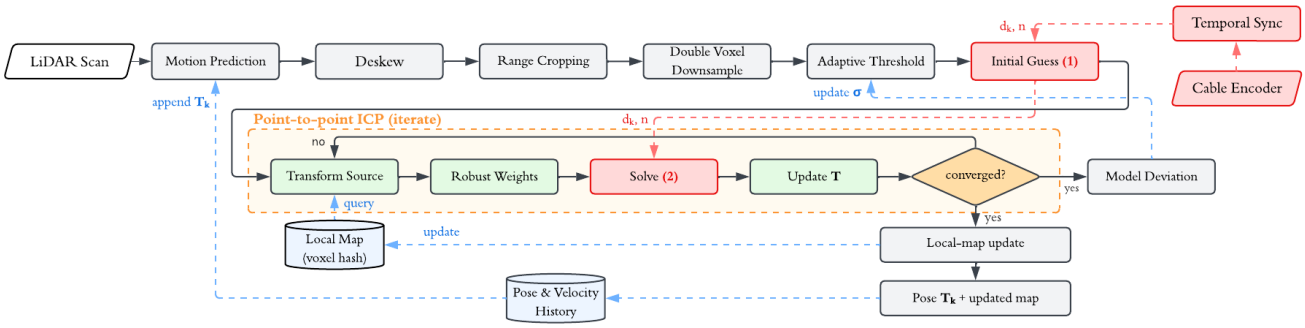


Fig. 4: KISS-ICP pipeline and anchor insertion points. The encoder-derived displacement d_k corrects the constant-velocity prediction before registration (1). During ICP optimization (2), the Gauss–Newton system is augmented with the axial cable constraint and tilt regularization.

Coupled to a 43 mm measuring pulley, the encoder provides a nominal linear resolution of approximately 0.23 mm per pulse.

IV. Cable-Encoder-Anchored KISS-ICP

The proposed formulation modifies the standard KISS-ICP pipeline at two stages. Before registration, the encoder-derived displacement corrects the constant-velocity prediction. During registration, the Gauss–Newton system is augmented with an axial cable constraint and a tilt regularization term motivated by the suspended deployment geometry. Both stages are enabled in the Anchored KISS-ICP configuration, as illustrated in Fig. 4.

A. Standard KISS-ICP Pipeline

KISS-ICP [11] is a LiDAR-only odometry method based on robust point-to-point ICP. For each incoming scan, a constant-velocity model derived from the two previous pose estimates predicts the sensor motion. This prediction is used for scan deskewing and registration initialization. The resulting cloud is range-filtered and voxel-downsampled before registration against a voxel-hashed local map.

Nearest-neighbor correspondences are optimized through Gauss–Newton iterations using a robust kernel. An adaptive threshold derived from the discrepancy between predicted and estimated motion controls correspondence rejection and the robust-kernel scale. After convergence, the registered scan is inserted into the local map and its pose is added to the trajectory history. The proposed modifications act only on the constant-velocity initial guess and the Gauss–Newton registration system.

B. Cable-Derived Axial Measurement

Throughout this paper, four related quantities are distinguished. Shaft depth is the independently measured collar-to-bottom distance; cable payout d_k is the encoder-derived released tether length; payload axial displacement is the sensor translation along the deployment axis; and reconstructed axial extent is the point-cloud span along that direction. Because tether elasticity, tension, and sensor FOV affect these relationships, these quantities are related but not interchangeable.

Let $\mathbf{T}_k = (\mathbf{R}_k, \mathbf{t}_k) \in SE(3)$ denote the LiDAR pose at scan k in the KISS-ICP world frame W. The world frame is initialized at the first LiDAR pose, such that $\mathbf{T}_0 = \mathbf{I}$.

The reel encoder is zeroed at the beginning of each descent and provides the cable payout d_k, with $d_0 = 0$. The unit vector $\mathbf{n}$ defines the positive deployment axis in W. The scalar product $\mathbf{n}^\top \mathbf{t}_k$ therefore represents the estimated axial displacement of the LiDAR along the deployment axis.

The cable residual is defined as

$$r_a(\mathbf{T}_k) = \mathbf{n}^\top \mathbf{t}_k - d_k. \tag{1}$$

i.e., the difference between the axial displacement estimated by KISS-ICP and that measured by the cable encoder.

Encoder samples are linearly interpolated at the LiDAR scan deskew reference time so that d_k and the motion-compensated point cloud refer to the same instant.

C. Strategy 1: Prediction Override

In standard KISS-ICP, the registration initial guess is obtained exclusively from the constant-velocity model. In the anchored formulation, the predicted orientation and lateral translation are preserved, while the encoder-derived displacement replaces the component along the deployment axis. For the predicted translation $\mathbf{t}_k^{\text{pred}}$,

$$\mathbf{t}_k^{\text{pred}} \leftarrow \mathbf{t}_k^{\text{pred}} + \left(d_k - \mathbf{n}^\top \mathbf{t}_k^{\text{pred}}\right) \mathbf{n}. \tag{2}$$

The correction acts only along $\mathbf{n}$. It therefore provides a depth-consistent initial guess for deskewing and registration without modifying the subsequent ICP procedure.

D. Strategy 2: Registration Constraints

The prediction override provides a depth-consistent initialization but does not prevent ICP from subsequently modifying the axial component of the pose. The encoder measurement is therefore retained during registration as an additional axial constraint. The suspended deployment also motivates a tilt regularization term that limits rotations orthogonal to the deployment axis.

At each ICP iteration, KISS-ICP estimates a pose increment $\delta\boldsymbol{\xi} = [\delta\boldsymbol{\rho}^\top \ \ \delta\boldsymbol{\phi}^\top]^\top \in \mathbb{R}^6$, where $\delta\boldsymbol{\rho}$ and $\delta\boldsymbol{\phi}$ are the translational and rotational increments, respectively. Using the left-multiplicative update $\mathbf{T}_k \leftarrow \mathrm{Exp}(\delta\boldsymbol{\xi}^\wedge)\mathbf{T}_k$, the Jacobian of the cable residual in (1) is $\mathbf{J}_a = [\mathbf{n}^\top \ (\mathbf{t}_k \times \mathbf{n})^\top]$.

For the tilt regularization, the current orientation relative to the initial pose is represented by $\boldsymbol{\phi}_k = \mathrm{Log}(\mathbf{R}_k)$, where $\mathrm{Log} : SO(3) \to \mathbb{R}^3$ denotes the rotation-vector logarithm. The projector $\mathbf{P} = \mathbf{I} - \mathbf{n}\mathbf{n}^\top$ removes rotation about the deployment axis, yielding the tilt residual $\mathbf{r}_{\mathrm{att}} = \mathbf{P}\boldsymbol{\phi}_k$. Under the small-angle approximation, its first-order Jacobian is $\mathbf{J}_{\mathrm{att}} = [\mathbf{0}_{3\times 3}\ \mathbf{P}]$.

The augmented Gauss–Newton system solved by the anchored configuration is therefore

$$\begin{aligned}\Big(\mathbf{H} + w_a \mathbf{J}_a^\top \mathbf{J}_a + w_{\mathrm{att}} \mathbf{J}_{\mathrm{att}}^\top \mathbf{J}_{\mathrm{att}}\Big)\delta\boldsymbol{\xi} \\ = -\Big(\mathbf{b} + w_a \mathbf{J}_a^\top r_a + w_{\mathrm{att}} \mathbf{J}_{\mathrm{att}}^\top \mathbf{r}_{\mathrm{att}}\Big),\end{aligned} \tag{3}$$

where $\mathbf{H}$ and $\mathbf{b}$ are the normal-equation terms of the original robust point-to-point ICP, w_a is the cable information weight, and w_{att} controls the tilt regularization. No IMU measurement enters this regularization; the prior penalizes tilt drift relative to the initial KISS-ICP orientation while leaving rotation about $\mathbf{n}$ unconstrained.

The tilt regularization weight w_{att} is a heuristic stabilization gain rather than a physically calibrated parameter. During preliminary development, values of 10, 50, and 100 were evaluated qualitatively based on the visible tilt deformation of the reconstructed shaft. Increasing the weight reduced this deformation, while no further relevant change was observed after $w_{att} = 100$. This value was therefore fixed for all reported experiments and was not selected using the reference depth or C2C metrics.

E. Tether Uncertainty

Cable payout only approximates payload displacement because elasticity, tension variation, and cable seating affect the deployed length. A static bench test under a 5-kgf load over tether lengths from 5 to 40 m showed approximately linear elongation ($R^2 = 0.997$), yielding an estimated axial stiffness of $EA \approx 39$ kN, equivalent to approximately 1.26 mm/m under the test load.

The measurement is therefore assigned a standard deviation σ_a that increases with deployed length, with the anchor weight defined as its scaled inverse variance:

$$\sigma_a(d_k) = \sigma_0 + \kappa\,|d_k|, \qquad w_a = \frac{s}{\sigma_a^2(d_k)}, \tag{4}$$

where σ_0 is the residual uncertainty near the initial reference, κ controls its growth with deployed length, and s is a dimensionless scale factor that adjusts the overall influence of the anchor relative to the LiDAR term without altering the uncertainty model. This formulation prevents the encoder-derived depth from being treated as an exact measurement.

In the experiments, $\sigma_0 = 0.02$ m, $\kappa = 0.002$ m/m, and $s = 1$. The value of κ was selected conservatively above the measured elastic slope to account for additional tension and seating effects, whereas σ_0 represents an empirical residual uncertainty near the initial reference; $s = 1$ applies no additional rescaling. Accordingly, σ_a increases from approximately 0.032 m at 6 m to 0.060 m at 20 m and 0.100 m at 40 m, reducing the cable influence with deployment depth. These values describe cable-cue uncertainty, not total mapping error, which must be evaluated experimentally in deeper shafts.

V. Experimental Setup

The experiments were performed in two shaft descents, denoted Shaft 1 and Shaft 2, located in the region of Ouro Preto, Minas Gerais, Brazil. All experiments were replayed offline from identical ROS 2 bag segments for all configurations. Only the descent phase was evaluated. The experimental setup, including the tripod, odometry cable and proposed sensing device is depicted in Fig. 5.



Fig. 5: Experimental setup: outdoor tripod with a custom-designed pulley that is attached to an encoder (left), and the proposed device entering the shaft bore (right).

FAST-LIO2 was initialized from the Mid-360 configuration provided with the original implementation. The LiDAR–IMU extrinsics, inertial-noise parameters, field of view, and temporal settings were retained, while scene-scale parameters were adapted to the confined geometry by increasing point retention and map resolution and restricting the blind, detection, and local-map ranges. LiDAR and IMU data shared a common ROS 2 time base, with additional temporal synchronization disabled.

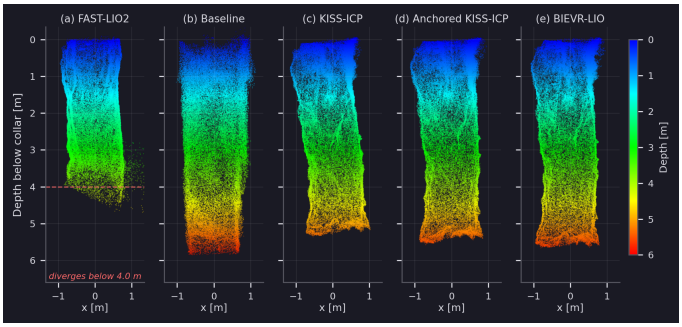
For comparison, the reconstructed point clouds were exported using a common voxel size of 0.02 m and identical range limits. Returns associated with the protective structure surrounding the LiDAR were removed through angular filtering before geometric evaluation. The same evaluation preprocessing was applied to all reconstructed clouds.

Shaft depth was measured manually from the collar using a laser single-point measurement device and served as the only reference independent of the estimation pipelines. Reconstructed axial extent was compared against these measurements. Cloud-to-cloud (C2C) distances were computed in CloudCompare after rigid, scale-locked alignment using local quadric modeling with (k=6) neighbors.

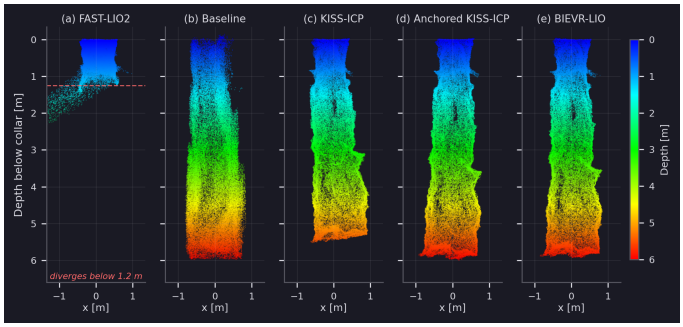
A. Evaluated Configurations

Five mapping configurations with different sources and uses of motion information were evaluated:

- **FAST-LIO2 [10].** The Mid-360 LiDAR and its built-in IMU were processed using the LiDAR–inertial FAST-LIO2 pipeline.



(a) Reconstructions of Shaft 1.



(b) Reconstructions of Shaft 2.

Fig. 6: The 3D reconstructions of two mining shafts: (a) Shaft 1, and (b) Shaft 2, both colored by depth using FAST-LIO2, attitude- and cable-referenced baseline, KISS-ICP, Anchored KISS-ICP, and BIEVR-LIO. All clouds have the same scale.

- **Attitude- and cable-referenced concatenation (baseline).** Each LiDAR scan was oriented using a Madgwick attitude estimate [19], with roll and pitch gravity-referenced and heading propagated without an absolute yaw reference. Axial translation was prescribed by the cable-encoder displacement, and no geometric scan registration was performed.
- **KISS-ICP [11].** The original LiDAR-only pipeline was applied without external motion constraints. The trajectory was estimated from scan-to-map registration using the standard constant-velocity prediction.
- **Anchored KISS-ICP (ours).** The modified pipeline described in Section IV combines the encoder-based prediction override in (2) with the axial and tilt registration constraints in (3). The tilt weight was set to $w_{\text{att}} = 100$ in all reported experiments.
- **BIEVR-LIO [12].** The same Mid-360 LiDAR and built-in IMU streams were processed with BIEVR-LIO.

VI. Results and Discussion

Figures 6a and 6b show the five reconstructions of Shaft 1 and Shaft 2 side by side, colored by depth. FAST-LIO2 did not preserve a single approximately axial shaft topology and was therefore retained only for qualitative comparison. Because axial extent and rigid C2C comparison assume such topology, these metrics were not applied to its reconstruction.

A. Reconstructed Axial Extent

Manual laser measurements from the shafts collar gave depths of 5.917 m and 5.951 m for Shafts 1 and 2, respectively. Table I compares these references with the reconstructed axial extent of the four configurations that preserved the shaft topology. LiDAR-only KISS-ICP showed the largest contraction in both sequences, with differences of (-9.21%) and (-6.75%). The anchored formulation reduced KISS-ICP's mean absolute difference from approximately 0.473 to 0.153 m, corresponding to a reduction of about 68%. BIEVR-LIO and the baseline achieved mean absolute differences of 0.136 and 0.170 m, respectively.

TABLE I: Reconstructed axial extent and error relative to manual laser depth measurement. Lowest absolute error per shaft in **bold**.

Shaft	Method	Extent [m]	Error [m]	Error [%]
1	Baseline	6.073	+0.156	+2.64
	KISS-ICP	5.372	−0.545	−9.21
	Anchored KISS-ICP	**5.627**	**-0.290**	**-4.90**
	BIEVR-LIO	**5.678**	**-0.239**	**-4.04**
2	Baseline	6.134	+0.183	+3.08
	KISS-ICP	5.549	−0.402	−6.75
	Anchored KISS-ICP	**5.935**	**-0.016**	**-0.26**
	BIEVR-LIO	**5.983**	**+0.032**	**+0.54**

B. Cross-Method Geometric Agreement

Cloud-to-cloud (C2C) distances were computed after rigid, scale-locked alignment to BIEVR-LIO using a 100% final-overlap setting. BIEVR-LIO was selected as the common reference because it preserved the shaft topology while remaining independent of the cable encoder. It is not treated as ground truth: C2C values are reference-dependent and are used only to characterize cross-method geometric agreement. Table II reports the resulting method-to-reference distances.

TABLE II: Cloud-to-cloud distances with respect to the BIEVR-LIO reconstruction. Best method highlighted in **bold**.

Shaft	Method	Mean [cm]	Median [cm]	P_{95} [cm]	≤ 2 cm [%]
1	Baseline	6.17	4.09	18.87	31.7
	KISS-ICP	5.87	4.09	17.33	31.7
	Anchored KISS-ICP	**5.64**	**3.90**	**16.63**	**32.3**
2	Baseline	5.01	3.29	15.43	36.3
	KISS-ICP	3.66	2.28	11.14	46.2
	Anchored KISS-ICP	**3.75**	**2.34**	**11.31**	**45.4**

BIEVR-LIO is the reference and FAST-LIO2 is excluded, as its reconstruction does not preserve the shaft topology.

All evaluated reconstructions remained at the centimeter scale, although the baseline showed the largest mean C2C distance in both shafts. Median distances ranged from 3.90 to 4.09 cm in Shaft 1 and from 2.28 to 3.29 cm in Shaft 2. The larger dispersion of the baseline is consistent with the visibly thicker wall surfaces in Figs. 6a and 6b. Because its heading is propagated without an absolute yaw reference, residual heading drift can superimpose wall segments at incorrect azimuths

while preserving their axial placement. Rotation about the shaft axis leaves a near-cylindrical surface largely unchanged, making C2C weakly sensitive to azimuthal misalignment.

C. Registration Repeatability and Geometric Degeneracy

To examine registration sensitivity, each of the six method–shaft pairs was aligned twice to the baseline reconstruction, using final-overlap values of 90% and 100%, yielding two transformations per pair. Between runs, translations along the shaft axis changed by 1.3–28.1 cm (mean 17.6 cm), and rotation about the axis changed by a mean of $4.2°$. Because the ICP routine also subsamples the clouds at each execution, these differences reflect the combined effect of the overlap setting and subsampling rather than overlap alone. Nevertheless, the resulting decimeter-level axial shifts and degree-level azimuthal changes are consistent with the weak observability associated with near-cylindrical geometry.

A more direct demonstration was obtained by aligning one pair (Anchored KISS-ICP and BIEVR-LIO, Shaft 2) through manually selected correspondences rather than automatic registration. The two solutions differ by $160°$ in rotation about the shaft axis, yet the manual alignment produced bidirectional mean C2C distances of $3.94/3.54$ cm (Anchored-to-BIEVR/BIEVR-to-Anchored) against 3.75 and 3.68 cm for the automatic one. Two alignments separated by nearly half a turn are therefore nearly indistinguishable under this metric.

D. Discussion and Limitations

No configuration was superior in all metrics. Under the evaluated configuration, FAST-LIO2 exhibited severe trajectory distortion, indicating that the presence of inertial aiding alone was insufficient to preserve the expected shaft topology in these sequences. Among the configurations that preserved a plausible reconstruction, the baseline reproduced axial extent well but generated less defined wall surfaces. LiDAR-only KISS-ICP contracted vertically and produced a curved axis in Shaft 2, while the anchored variant mitigated both effects through the combined axial and tilt constraints. BIEVR-LIO achieved comparable global consistency through a different LiDAR–inertial formulation.

VII. Conclusion and Future Research

This paper compared five mapping strategies for cable-lowered historical shaft inspection and introduced an Anchored KISS-ICP formulation combining an encoder-based prediction override, a soft axial constraint, and a tilt prior. FAST-LIO2 produced severe trajectory distortion, while LiDAR-only KISS-ICP showed the largest axial contraction among the remaining methods. Cable anchoring reduced KISS-ICP's mean absolute extent error from 0.473 to 0.153 m. Registration was least repeatable in axial translation and shaft-axis rotation, consistent with the near-cylindrical geometry. The contrasting behavior of the two LiDAR–inertial pipelines further shows that auxiliary sensing alone does not eliminate degeneracy; robustness also depends on how motion and geometric information are represented and fused.

Future work will evaluate deeper and more diverse shafts using survey-grade references, investigate ascent–descent consistency, isolate the individual anchor terms, refine the tether model, and improve sensor placement and orientation.

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